R4 JR Round: Jeopardy

Subjects: Computer Science, Mathematics, Physics, Biology, Environmental Studies, Chemistry, Mystery

Part 1: Questions

Computer Science

1. Which logic gate gives an output of 1 only when both of its inputs are 1?

A) OR
B) NOT
C) XOR
D) AND

2. What is the output of this Python code?

```
x = 5
x += 3
x *= 2
print(x)
```

A) 13
B) 16
C) 18
D) 10

3. What is the result of adding the binary numbers 1011 and 0110?

A) 10001
B) 10011
C) 11001
D) 10101

4. What is the hexadecimal equivalent of the 4 bit binary number 1111?

A) 1
B) 4
C) F
D) 16

5. In pseudocode, a check reads: IF (Age >= 18) OR (HasID = TRUE AND IsVip = TRUE). If Age = 20, HasID = FALSE, and IsVip = FALSE are inputs given, what happens when code is run?

A) Error is generated as inputs are out of range
B) Conditions are not satisfied
C) Conditions are satisfied
D) Wrong syntax

6. How many stars (*) does this pseudocode print?

```
For i from 1 to 4:
    For j from 1 to i:
        Print "*"
```

A) 4
B) 10
C) 8
D) 16

Mathematics

1. Evaluate $(-3)^2 - 3^2$

A) 0
B) 18
C) −18
D) 9

2. A bag contains 3 red balls and 7 blue balls. A ball is drawn at random and not replaced. A second ball is drawn. What is the probability that the second ball is red, given that the first ball drawn was red?

A) 3/10
B) 2/9
C) 3/9
D) 2/10

3. Solve for $\sqrt{(x^2)} = 9$

A) +9, −9
B) +9
C) −9
D) Neither of the above

4. A 13 m ladder leans against a vertical wall, with its foot 5 m from the base of the wall. How high up the wall does the ladder reach?

A) 8 m
B) 12 m
C) 14 m
D) 18 m

5. For what values of k does the equation $2x^2 + kx + 8 = 0$ have equal roots?

A) $k = \pm 4$
B) $k = \pm 8$
C) $k = 4$
D) $k = 8$

6. Find the compound interest on Rs. 10,000 for 2 years at 10% per annum, compounded yearly.

A) Rs. 2,100
B) Rs. 2,000
C) Rs. 2,200
D) Rs. 12,100

Physics

1. A 12 V battery is connected across a 4 Ω resistor. What current flows through the resistor?

A) 3 A
B) 8 A
C) 16 A
D) 48 A

2. A car travels 10 km East, turns around, and travels 10 km West in 20 minutes. What is the car's average velocity in km/h?

A) −60 km/h
B) 0 km/h
C) 1 km/h
D) 60 km/h

3. A stone is dropped from rest. What is its speed after 3 seconds of free fall?

A) 10 m/s
B) 20 m/s
C) 30 m/s
D) 45 m/s

4. A 1000 W heater is used to warm 2 kg of water by 10 °C. How long does it take, assuming no heat is lost? (specific heat capacity of water = 4200 J/kg°C)

A) 42 s
B) 168 s
C) 84 s
D) 8.4 s

5. Two resistors of 4 Ω and 6 Ω are connected in parallel across an ideal 12 V battery. Which resistor dissipates more power?

A) 4 Ω
B) 6 Ω
C) Same for both resistors as they are in parallel
D) Resistors do not dissipate power

6. An object is placed 15 cm in front of a diverging (concave) lens with a focal length of 10 cm. Is the image real or virtual, and is it inverted or upright?

A) Real, Inverted
B) Real, Upright
C) Virtual, Inverted
D) Virtual, Upright

Biology

1. Which of the following describes the bonds found in one molecule of water?

W) Hydrogen
X) Nonpolar covalent
Y) Polar covalent
Z) Ionic

2. A protein's tertiary structure is most strongly influenced by:

W) ATP synthetase (pronounced: SIN the tase)
X) imbedding in the plasma membrane
Y) ionic and hydrogen bonds between R groups
Z) the quaternary structure

3. Excess secretion from this gland can cause a person to be thin, hyperactive, always hungry, and irritable.

W) adrenal cortex
X) thyroid
Y) anterior pituitary
Z) adrenal medulla

4. Transfer RNA has all of the following characteristics, EXCEPT which of the following?

W) It travels from the nucleus to the cytoplasm in eukaryotes
X) It is attached to an amino acid at the 5 prime end
Y) Its three dimensional structure is roughly L shaped
Z) It can be used repeatedly

5. What is water loss through the stomata of a plant called:

W) condensation
X) liquidation
Y) transpiration
Z) exchange

6. In early experiments in the mid 1900's, the source and fate of oxygen involved in photosynthesis was mostly determined through experiments that used:

W) animals in closed containers that would die unless a plant was introduced into the container
X) a heavy isotope of oxygen as a tracer
Y) fluorescent labeling of various enzymes and organic products
Z) simple accounting of the atomic mass of all the reactants and products

Environmental Studies

1. What forms when streams enter bodies of standing water, undergo an abrupt loss of velocity, and drop their load of sediment?

W) Alluvial fans
X) Taluscones
Y) Vasnes
Z) Deltas

2. Geothermal power plants use which of the following to generate electricity:

W) coal
X) steam
Y) sunlight
Z) pressurized water

3. Named in honor of the English Mathematician who explained them, what is the name for the thermal circulation consisting of rising air near the equator and sinking air near 30° latitude:

W) Rossby wave
X) Ferrel cell
Y) Hadley cell
Z) Christopher cell

4. A major function of a refractory is

W) the conduction of heat
X) for heat insulation
Y) as a source of essential reactants for smelting
Z) as a source of flux material for slag formation

5. Chemical weathering of sodium and plagioclase feldspar result in hydrous aluminous silicates otherwise known as?

W) Evaporites
X) Clay minerals
Y) Precipitates
Z) Sheet silicates

6. All of the following are formed by deposition EXCEPT:

W) sea stack
X) beach
Y) sandbar
Z) spit

Chemistry

1. Sulfur forms the following compounds: SO_2, SCl_2, and SO_3^{2-}. Which form of hybridization is NOT represented by these molecules?

W) sp
X) sp^2
Y) sp^3
Z) None of these

2. Which of the following is least related to the strength of a covalent bond?

W) vibrational frequency
X) bond order
Y) bond length
Z) bond direction

3. Which of the following scientists hypothesized that it would be possible for particles to have wave like properties? Was it:

W) Planck
X) Heisenberg
Y) Einstein
Z) de Broglie

4. An amine is a class of compounds that contains a:

W) sulfur and hydrogen group
X) nitrogen and oxygen group
Y) nitrogen and hydrogen group
Z) carbon and sulfur group

5. Out of the following choices, which is the correct name for N_2O_3:

W) dinitrogen oxide
X) dinitrogen dioxide
Y) dinitrogen trioxide
Z) dinitrogen tetroxide

6. Which of the following is the theoretical end of a titration:

W) end point
X) indicator point
Y) equivalence point
Z) standardization point

Mystery

1. Known as the “Human Computer” of India, she set a world record by mentally multiplying two 13 digit numbers in just 28 seconds. Name her.

A) Janaki Ammal
B) Shakuntala Devi
C) Kalpana Chawla
D) Asima Chatterjee

2. Who was the first Indian origin woman to travel into space aboard the Space Shuttle Columbia in 1997?

A) Sunita Williams
B) Shawna Pandya
C) Kalpana Chawla
D) Sirisha Bandla

3. Born in Lahore, this Pakistani American astrophysicist served as the Dean of Science at MIT and helped detect gravitational waves. Name her.

A) Dr. Nergis Mavalvala
B) Dr. Tasneem Zehra Husain
C) Dr. Perween Rahman
D) Dr. Abha Sur

4. Which pioneering botanist cross bred sugarcane varieties in the 1930s to make them sweeter and suit India’s tropical climate?

A) Asima Chatterjee
B) Anna Mani
C) Archana Sharma
D) Janaki Ammal

5. Who holds the distinction of being the first Pakistani woman to earn a PhD in Theoretical Physics / String Theory?

A) Dr. Nergis Mavalvala
B) Dr. Tasneem Zehra Husain
C) Dr. Samina Shah
D) Dr. Azra Raza

6. She was the first woman to earn a Doctorate of Science from an Indian university (1944) for developing plant based drugs to treat malaria. Who was she?

A) Kamala Sohonie
B) Asima Chatterjee
C) Rajeshwari Chatterjee
D) Raman Parimala

Part 2: Answers

Computer Science

1. D) AND
2. B) 16
3. A) 10001
4. C) F
5. C) Conditions are satisfied
6. B) 10

Mathematics

1. A) 0
2. B) 2/9
3. A) +9, −9
4. B) 12 m
5. B) k = ±8
6. A) Rs. 2,100

The handwritten solutions for the Mathematics questions are on the next page.

Date:
M T W T F S S

SOLUTIONS FOR JEOPARDY:

Q1:

$$(-3)^2 - 3^2$$
$$9 - 9$$
$$= 0$$

Q2:

$$P(\text{red}_2 / \text{red}_1) = \frac{3-1}{(3+7)-1} = \frac{2}{9}$$

Total balls = 3 + 7

Total remaining balls = 2 + 7 = 9

Q3: $\sqrt{x^2} = |x| = 9$

$$x = \pm 9$$

Q4: $h = \sqrt{13^2 - 5^2}$

$$h = \sqrt{169 - 25}$$
$$h = \sqrt{144} = 12\,m$$

Q5. $b^2 - 4ac = 0$

$$k^2 - (4)(2)(8) = 0$$
$$k^2 - 64 = 0$$
$$k = \pm 8$$

Q6. $CI = 10000\left(1 + \frac{10}{100}\right)^2 - 10000$

$$= 2100\ Rs$$

CamScanner

Part 2: Answers (continued)

Physics

1. A) 3 A
2. B) 0 km/h
3. C) 30 m/s
4. C) 84 s
5. A) 4 Ω
6. D) Virtual, Upright

Biology

1. Y) Polar covalent
2. Y) ionic and hydrogen bonds between R groups
3. X) thyroid
4. X) It is attached to an amino acid at the 5 prime end
5. Y) transpiration
6. X) a heavy isotope of oxygen as a tracer

Environmental Studies

1. Z) Deltas
2. X) steam
3. Y) Hadley cell
4. X) for heat insulation
5. X) Clay minerals
6. W) sea stack

Chemistry

1. W) sp
2. Z) bond direction
3. Z) de Broglie
4. Y) nitrogen and hydrogen group
5. Y) dinitrogen trioxide
6. Y) equivalence point

Mystery

1. B) Shakuntala Devi
2. C) Kalpana Chawla
3. A) Dr. Nergis Mavalvala
4. D) Janaki Ammal
5. B) Dr. Tasneem Zehra Husain
6. B) Asima Chatterjee